

Smart Compression Knee Sleeve for Post- Injury Rehabilitation

CREST Gold Award Report

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Abstract

Anterior Cruciate Ligament (ACL) injuries are a serious health issue, comprising 50% of all knee ligament injuries. Improper recovery protocols contribute to an alarming secondary injury rate as high as 23%. This project aimed to develop and validate a reliable system for quantitatively assessing post-operative physiotherapy. The goal was to provide objective data on patient recovery progress to improve rehabilitation outcomes.

A prototype system was developed to measure key biomechanical metrics. Data was processed using a custom Python pipeline. The study compared three participant groups: a baseline of healthy individuals (n=5) and two injured participants (n=2). We analyzed both their recovery and their unaffected leg. Four key metrics were calculated: Knee Flexion Range (deg), Sway Path (deg), 95% Ellipse Area (degZ), and Angular Velocity RMS (deg/s).

The initial investigation successfully established the system's reliability, addressing a primary research question. Quantitative analysis revealed distinct differences between the participant groups. The recovering legs of injured participants showed altered performance across the four key metrics. This was evident when compared to both their unaffected leg and the healthy baseline group, highlighting specific recovery deficits.

We conclude that the developed prototype is a reliable tool for monitoring ACL rehabilitation. It successfully quantifies recovery and identifies biomechanical asymmetries between limbs. This objective data is crucial for evaluating physiotherapy protocols. The system helps restore a patient's range of motion and proprioception.

This technology has significant potential to improve clinical practice. It provides physiotherapists with data-driven insights to guide personalized rehabilitation. This could improve patient outcomes and reduce secondary ACL injury rates. The system offers an accessible tool for enhancing post-operative care.

Introduction

Background Information

The human knee is a biomechanical marvel, yet its complexity makes it highly susceptible to injury. Knee injuries are serious health issues, accounting for 41% of all sports injuries. Among these, Anterior Cruciate Ligament (ACL) injuries are very common, comprising roughly 50% of all knee ligament injuries. Following surgery, physiotherapy is critical for recovery. However, improper recovery protocols contribute to an alarming secondary ACL injury rate, as high as 23%. A major problem is how recovery is monitored. Many clinics rely on subjective methods like an “eye-test”, which can miss subtle movement errors. Conversely, objective “gold standard” systems are extremely expensive and inaccessible. This creates a vast measurement gap, especially in India.

Project Aim

This research aims to bridge this measurement gap using modern wearable technology. The project’s primary aim was to build and test a prototype “Smart Compression Sleeve”. This low-cost device is designed to provide objective, quantitative data to improve monitoring of post-operative ACL rehabilitation and patient outcomes.

Objectives

To achieve our aim, the project was guided by the following specific objectives:

1. To implement and validate a wearable system. The system uses dual- IMU and flex sensors to transmit stable sensor data for analysis via Bluetooth.
2. To assess the consistency and reliability of the data from the wearable system. This was done by having a non-injured participant repeat the same movement tasks multiple times.
3. To determine if the system can detect clear, quantitative differences in balance and knee movement. We compared data from participants with knee injuries to healthy participants.
4. To extract and compare key biomechanical metrics. Metrics like Sway Path and Ellipse Area were used to quantify the differences between the study groups.

Significance

This research holds significant potential to improve clinical physiotherapy. Current subjective assessments create ambiguity in patient evaluation.

Failing to meet objective discharge criteria is associated with a four-times higher risk of graft rupture. Our project directly addresses this issue by offering an accessible, data-driven tool. The system provides an alternative to expensive motion capture labs. This is particularly crucial in settings like India, where clinics often rely entirely on observation-based methods. The “Smart Compression Sleeve” empowers therapists with objective data. This helps in accurately restoring a patient’s range of motion and proprioception. By quantifying movement patterns, the system helps identify high-speed errors that the human eye might miss. This allows for more effective, personalized rehabilitation programmes. The ultimate impact is a potential reduction in the high rate of secondary ACL injuries. It offers a practical solution to enhance post-operative care, making objective biomechanical analysis more accessible and affordable for clinics everywhere.

Literature Review

Current State of Research

Knee injuries are a significant and prevalent issue in sports medicine and public health. They constitute a large portion of all sports-related injuries, with some studies reporting rates as high as 41% (Golberg et al., 2023). Within this category, injuries to the Anterior Cruciate Ligament (ACL) are particularly common and debilitating. They account for approximately half of all knee ligament injuries (Tan et al., 2023). The consequences of an ACL tear are severe. They often necessitate surgical reconstruction followed by a lengthy and complex rehabilitation process. The median return-to-sport time can often exceed one year, which has profound implications for athletes (Wu et al., 2023). Beyond physical recovery, these injuries carry significant personal and economic costs for patients and healthcare systems (Guan et al., 2024).

Effective post-operative physiotherapy is paramount for successful recovery. The primary clinical goals are to restore the knee's full range of motion (ROM) and rebuild muscle strength. Equally important is regaining proprioception, which is the body's intrinsic sense of its position in space (Porciuncula et al., 2018). Unfortunately, failures in rehabilitation are common. Improper or poorly monitored recovery protocols contribute to a high rate of secondary ACL injuries. This rate can be as high as 23% in young athletes, representing a major clinical challenge (Tan et al., 2023). A critical factor in this high re-injury rate is the methodology used to assess recovery progress. Traditional clinical assessment often relies on subjective observation or manual tools like goniometers. These methods are prone to error and may fail to detect subtle biomechanical deficits, leading to premature clearance for return to sport.

Key Studies and Findings

Recent academic literature has increasingly focused on the limitations of subjective assessment and the potential of technology to provide objective metrics. A pivotal scoping review by Marques et al. (2022) explored the use of wearable technology to identify between-limb differences following ACL reconstruction. Their work highlights a crucial finding: failing to meet objective discharge criteria is associated with a four-times higher risk of graft rupture. This underscores the urgent need for quantitative, data-driven approaches in physiotherapy.

The field has seen a surge in the development and validation of wearable sensors for rehabilitation. A focused review by Porciuncula et al. (2018) charted the technological and clinical advances in wearable movement sensors for rehabilitation. It established their utility in providing continuous, objective feedback outside of a traditional lab setting. This trend is further supported by a systematic review from Guan et al. (2024), which assessed the validity of wearable sensors for rehabilitation following total

knee arthroplasty. Their findings confirm that wearables can provide reliable data, paving the way for their broader clinical adoption.

Specifically for ACL injuries, research has demonstrated the value of portable sensing technologies. A scoping review by Tan et al. (2023) examined the landscape of portable sensing for both ACL injury prevention and rehabilitation assessment. This work confirms that such devices can capture critical biomechanical data in real-world environments. Similarly, a systematic review by Wu et al. (2023) focused on monitoring external workload with wearable technology after ACL reconstruction. Their findings suggest wearables are effective tools for managing patient activity levels and ensuring a safe, progressive return to sport. These studies collectively build a strong case for integrating technology into standard ACL recovery protocols.

Further research has focused on comparing these emerging technologies to established standards. Milosevic et al. (2020) conducted an experimental comparison between consumer-grade systems like the Kinect and wearable inertial sensors for home-based motor rehabilitation. Their work illustrates the trade-offs between cost, accuracy, and accessibility, concluding that wearable sensors offer a powerful and practical solution for many applications. More recently, studies have focused on creating integrated, user-friendly systems. For example, Lee et al. (2024) conducted a validation study on an IoT-enabled knee sleeve for home rehabilitation, demonstrating the feasibility of such a device. This research validates the concept of a smart garment that can capture and transmit data for remote monitoring.

Research Gaps

Despite these advancements, a significant gap remains between sophisticated laboratory-based analysis and typical clinical practice. The “gold standard” for motion analysis involves expensive, complex optical motion capture systems. These systems require dedicated laboratory space and highly trained technicians, making them inaccessible for most clinics (Milosevic et al., 2020). At the other end of the spectrum is the subjective “eye-test,” which lacks objectivity and reliability (Marques et al., 2022). This creates a vast measurement and accessibility gap. While the literature validates the effectiveness of wearable sensors (Golberg et al., 2023; Tan et al., 2023), there is a persistent need for low-cost, clinically validated systems that can be easily deployed in diverse healthcare settings. This gap is particularly pronounced in developing countries like India, where financial constraints often limit access to advanced medical technology (Lee et al., 2024).

Moreover, while many studies validate specific sensors or algorithms, fewer focus on the end-to-end implementation and validation of a complete, low-cost system that is ready for clinical trials. There is a need for more research that not only develops the hardware but also validates its data against patient outcomes in a real-world context.

Relevance to Project

This project is positioned directly within this identified research gap. Its primary aim is to develop and test a low-cost “Smart Compression Sleeve” that integrates inertial measurement units (IMUs) and flex sensors. This work builds directly upon the foundational research confirming the utility of wearable sensors for rehabilitation (Porciuncula et al., 2018; Golberg et al., 2023). Our approach is similar to the IoT-enabled sleeve validated by Lee et al. (2024), but it focuses on a specific combination of low-cost components to maximize accessibility.

The project seeks to provide a practical solution to the problems highlighted by Marques et al. (2022). By offering objective, quantitative data on knee movement and balance, our system aims to reduce the ambiguity of subjective assessments. It will provide clinicians with the data needed to make informed decisions about patient progress and readiness for return to sport. By addressing the challenge of monitoring patient workload (Wu et al., 2023) and identifying limb asymmetries (Marques et al., 2022), this research aims to bridge the gap between expensive lab equipment and inadequate clinical observation. Ultimately, this project contributes to a tangible, accessible tool designed to improve physiotherapy outcomes and reduce the alarmingly high rates of secondary ACL injury.

Preparation

Planning

The project began with a thorough needs assessment. I identified a gap in objective ACL recovery monitoring. The primary goal was to create a data-driven tool for physiotherapists. I set clear objectives for the device's functionality. This included accurate knee flexion and angle measurement. The project scope was carefully defined. It covered hardware design, software programming, and data analysis. I established key milestones for each stage. These included prototype completion and successful data collection. Resource planning involved sourcing IMUs and flex sensors. I also allocated time for learning Python for data processing. A risk assessment was conducted to identify potential issues. These included sensor inaccuracy and participant recruitment delays. Success was defined by a functional prototype. It had to provide reliable and valid biomechanical data.

Resources and Training

I undertook significant self-training to acquire the necessary skills. I studied wearable sensor systems and biomechanical analysis. This included learning about IMUs and flex sensors. I focused on sensor placement and data fusion algorithms. This knowledge was crucial for ensuring data quality. I also developed skills in experimental design. This helped me create a robust data collection protocol. I learned statistical methods like the coefficient of variation. This was essential for analysing the reliability of my measurements. I gained practical experience with embedded systems. This project taught me how to use robotics to solve real-world problems. This preparation was fundamental to the project's success.

Timeline

The project was structured into four distinct phases over five months. The initial phase focused on literature review and component procurement. This stage proceeded as planned, setting a strong foundation. The second phase involved hardware prototyping and software development. This phase required an extension of two weeks. This was due to unexpected challenges in sensor integration. The third phase was dedicated to data collection. It was completed efficiently ahead of schedule. This was possible because of a well-defined movement protocol. The final phase involved data analysis and report writing. This was completed within the allocated timeframe. The initial delay in hardware development did not impact the final deadline. This was due to the accelerated data collection process. Effective time management was crucial for the project's success.

Methodology

Approach

This project adopted a quantitative, experimental research approach. The primary goal was to develop and validate a novel technological solution to a clear clinical problem. This approach was chosen because the research questions required objective, numerical data to be answered effectively. The research strategy was executed in two distinct phases. The first phase involved the design, fabrication, and programming of a low-cost, wearable “Smart Compression Sleeve” prototype. This phase focused on engineering and system integration. The second phase involved a human participant study to validate the prototype’s performance. This validation phase was designed to assess the system's reliability and its ability to differentiate between injured and non-injured individuals. This structured, two-phase approach allowed for the systematic development of the hardware, followed by a rigorous test of its practical application in a controlled setting. The overall methodology was designed to directly address the identified measurement gap between expensive laboratory systems and subjective clinical assessments. It aimed to provide empirical evidence for the device's utility in physiotherapy monitoring.

Experimental Design

The study employed a comparative cross-sectional design. This design was chosen to quantitatively compare biomechanical performance between different groups at a single point in time. A total of nine participants were recruited and divided into two primary cohorts. The first was a Baseline Group (n=5) consisting of healthy individuals with no history of knee injury. This group served as the control. The second was an Injured Group (n=4) comprising participants actively undergoing rehabilitation for knee injuries. The independent variable in this study was the participant's knee condition (healthy vs. injured). The dependent variables were the four key biomechanical metrics calculated from the sensor data. These metrics included Knee Flexion Range, Sway Path, 95% Ellipse Area, and Angular Velocity RMS. To ensure consistency and control for procedural variability, all participants performed the exact same set of six standardized movement tasks. These tasks were designed to challenge balance, strength, and coordination.

Materials and Equipment

All hardware components were selected to balance performance, portability, and low cost.

Hardware:

- **Microcontroller:** An ESP32 Dev Kit served as the central processing unit. It was chosen for its sufficient processing power and integrated Bluetooth Low Energy (BLE) radio, which simplifies the wireless design.
- **Inertial Measurement Units (IMUs):** Two Adafruit BNO055 9-axis sensors were used to measure 3D motion. Each BNO055 contains an accelerometer, gyroscope, and magnetometer. Its key feature is a built-in fusion algorithm that provides a stable orientation output, correcting for gyroscope drift.
- **Flex Sensor:** A Spectra flex sensor provided a direct measurement of the knee's bending angle. This sensor acts as a variable resistor, with its resistance changing as it is bent.
- **Power System:** A standard 3.7 Volt Lithium-Ion (Li-ion) battery was used to power the entire wearable system.
- **Fabrication Materials:** A standard nylon compression sleeve was used as the wearable base. Hot glue and strong adhesive were used to securely affix all electronic components to the sleeve.

Software and Applications:

- **Data Logging:** The nRF Connect for Mobile application was used on a standard smartphone to receive and log the transmitted BLE data packets into CSV files.
- **Data Analysis:** A custom data analysis pipeline was developed in Python. This pipeline utilized standard scientific libraries for data processing, filtering, and metric calculation.

Procedures

The methodology followed a structured, multi-stage process from system fabrication to final data analysis.

1. System Fabrication and Assembly: The prototype was constructed by carefully integrating the electronic components onto the nylon compression sleeve. The two BNO055 IMUs were connected to the ESP32's I2C data bus. The flex sensor was connected to one of the ESP32's analog-to-digital converter (ADC) pins. Figure 1 shows the architecture of the system.

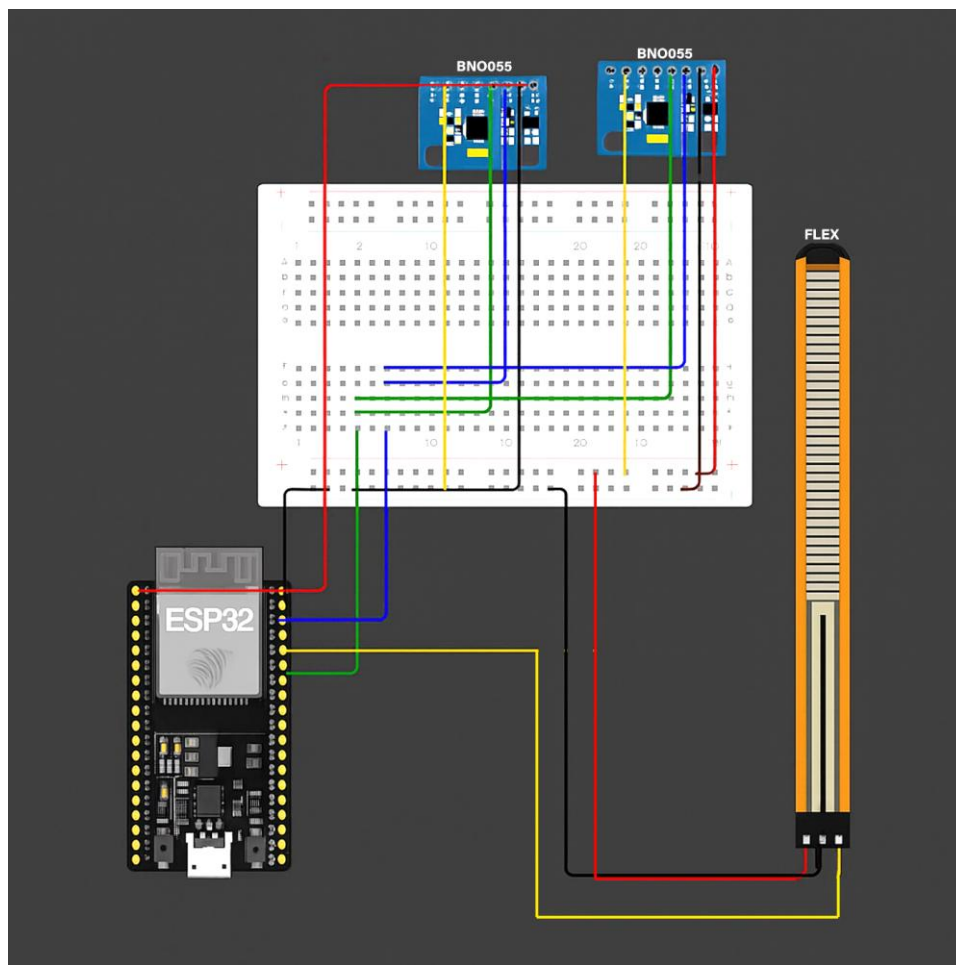


Figure 1: System architecture diagram. The diagram shows the flow of data. It starts from the BNO055 IMUs and the Flex Sensor. Data goes to the ESP32 microcontroller. The ESP32 then transmits data.

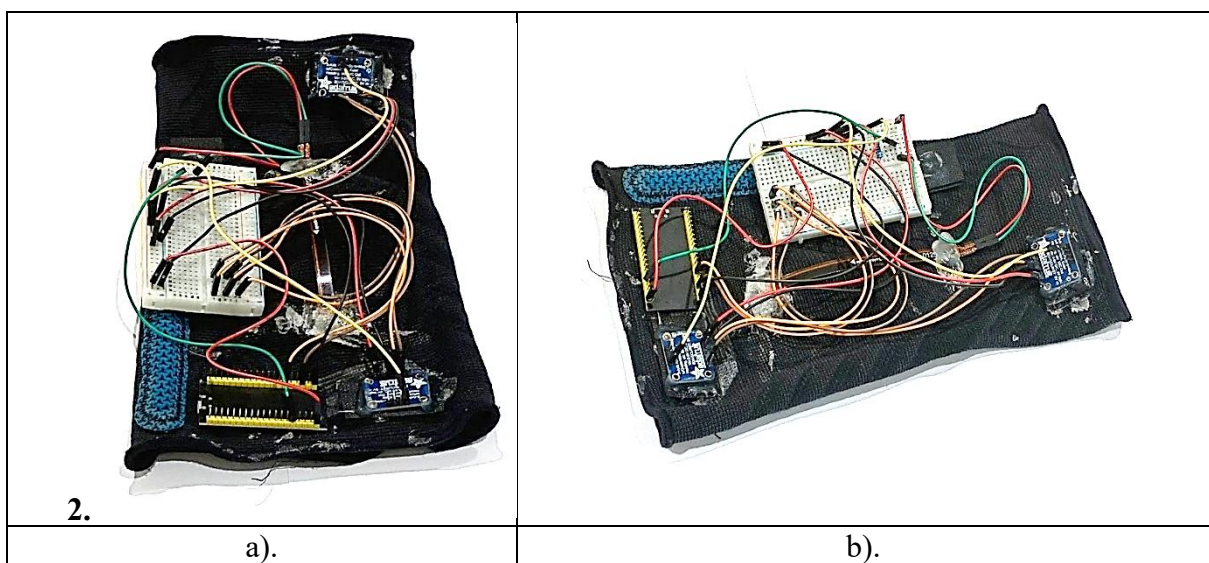


Figure 2: The "Smart Compression Sleeve" prototype. This photograph shows the placement of the two BNO055 IMUs (above and below the knee). The ESP32 microcontroller and flex sensor are also visible, all mounted on the nylon sleeve.

Figure 2 shows a photograph of an electronic circuit prototype assembled on a piece of black fabric, likely for a wearable device. The setup includes a central white solderless breadboard with numerous multi-colored jumper wires connecting various components. Several modules, including two small blue PCBs and a larger black PCB with yellow pin headers, are attached to the fabric base, seemingly with hot glue. Sensor placement was critical for data accuracy. IMU 1 was secured 10 cm above the patella to track thigh movement. IMU 2 was secured 10 cm below the patella to track shank movement. The flex sensor was positioned directly across the lateral side of the knee joint. All components were firmly attached using hot glue and adhesive to prevent any movement during trials, ensuring data integrity.

3. Data Collection Protocol: Each data collection session with a participant lasted approximately 20 minutes. The procedure began by fitting the “Smart Compression Sleeve” onto the participant's designated leg. Using the nRF Connect app on a smartphone, a BLE connection was established with the device. The participant was then instructed to perform a series of six specific movement tests. These tests were: 10m Jog, Double Step Stair Climbing, Triple Step Stair Climbing, Up Down Stair Climbing, Leg Sway, and Knee Up.

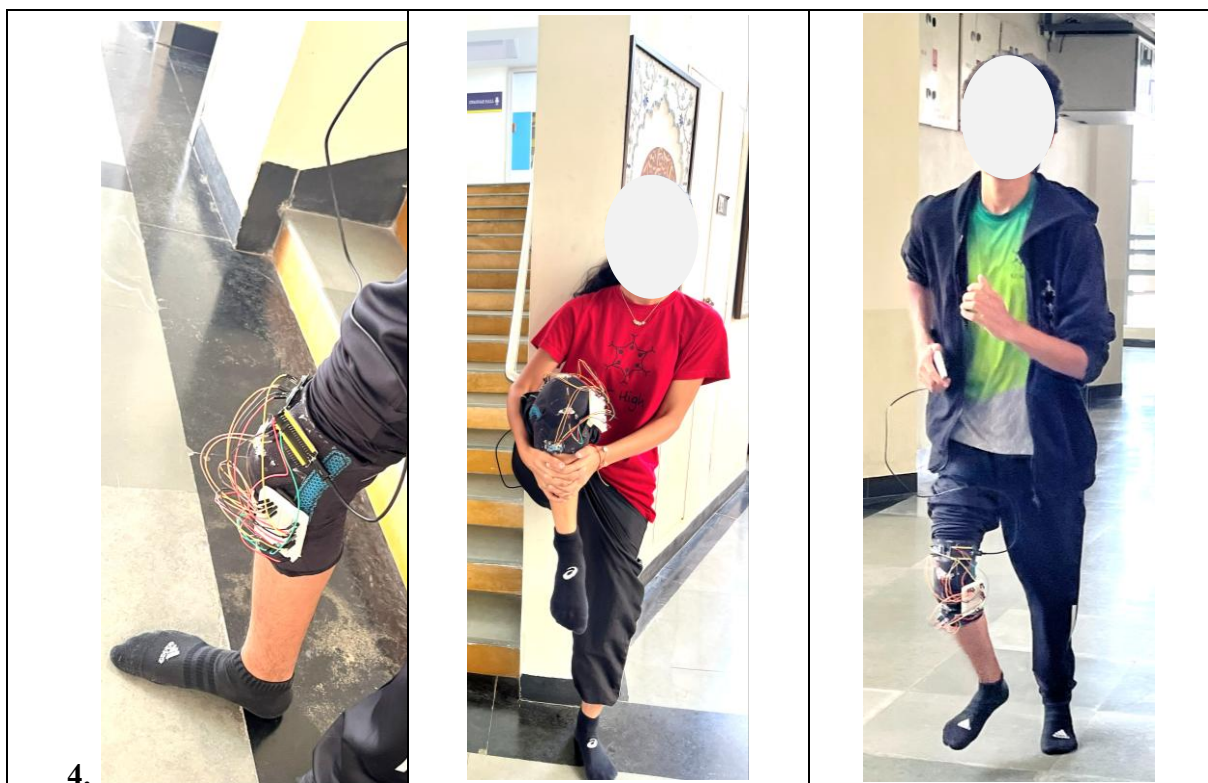


Figure 3: Experimental setup in progress. A participant is shown wearing the smart sleeve while performing a movement task. The researcher (not pictured) monitors the data logging on a nearby system.

5. Data Analysis Pipeline: All recorded data was processed offline using the custom Python pipeline. This pipeline executed a rigorous multi-stage process.

- **Pre-processing:** First, raw log files from the smartphone were cleaned to extract the valid 20-column sensor data format, removing any extraneous text or connection artefacts.
- **Sensor Fusion:** A Madgwick fusion algorithm was implemented. This algorithm fused the raw 9-axis data (accelerometer, gyroscope, magnetometer) from each IMU. It converted these readings into stable 3D orientation data, specifically Euler angles (roll, pitch, yaw).
- **Filtering:** A 4th-order, 6Hz Butterworth low-pass filter was applied to all processed sensor signals. This included the Euler angles and the raw flex sensor data. This step was crucial for removing high-frequency noise and isolating the true physiological movement signals.
- **Metric Calculation:** Finally, four key biomechanical metrics were calculated from the clean, filtered data. Knee Flexion Range (deg) was derived from the flex sensor. Sway Path (deg) and 95% Ellipse Area (degZ) were calculated from the thigh IMU's roll-pitch trajectory to quantify postural stability. Angular Velocity RMS (deg/s) was calculated from the gyroscope data to quantify movement smoothness or shakiness.

Safety and Ethical Considerations

This study was conducted with the highest commitment to participant safety and ethical principles.

Ethical Conduct: Strict ethical protocols were followed throughout the research. Prior to participation, the study's purpose, procedures, potential risks, and benefits were clearly explained to every individual. Informed verbal consent was obtained from all participants. For participants who were minors, consent was also obtained from their legal guardians. To protect participant confidentiality, all collected data was fully anonymized at the source. Unique identifiers were used for analysis, and no personally identifiable information was stored with the sensor data. This ensured that individual privacy was maintained at all stages of the research process, from data collection to publication. Participants were also informed of their right to withdraw from the study at any time without any penalty.

Safety Measures: The physical safety of participants was a top priority. The wearable device itself was designed to be non-invasive and safe. It was powered by a low-voltage 3.7V battery, which eliminated any significant electrical risk to the user. During the fabrication process, care was taken to ensure there were no sharp edges or components that could cause discomfort. All movement trials were conducted in a clear, open space free from obstacles to minimize the risk of trips or falls. A researcher supervised every session closely, especially for participants in the injured group, to aid if needed. The compression sleeve was checked for a secure but comfortable fit to

ensure it did not restrict circulation or movement. The chosen activities, while challenging, were common physiotherapy exercises and were performed within the comfortable limits of each participant.

Results

Data Presentation and Analysis

This section presents the quantitative findings from the experimental study. All data was collected and processed using the custom Python pipeline as detailed in the methodology. The analysis focuses on a direct comparison between three distinct participant groups. These groups are: the Baseline group (n=5), comprising healthy participants with no history of knee injury; the Injured LL group (n=2), representing the unaffected or “good” leg of two injured participants; and the Injured RL group (n=2), representing the recovering or “bad” leg of the same two injured participants. The results are organized logically to first establish the foundational reliability of the measurement system itself, thereby addressing Research Question 1 (RQ1). Following this validation, the section presents the primary findings related to patient biomechanical performance, directly addressing the core objective of Research Question 2 (RQ2). Four key biomechanical metrics were calculated for every trial conducted by each participant: Knee Flexion Range (in degrees), Sway Path (in degrees), 95% Ellipse Area (in degrees squared), and Angular Velocity RMS (in degrees per second). These metrics were specifically chosen to provide a comprehensive, quantitative assessment of both joint motion and neuromuscular control, which are critical components of a successful post-operative recovery.

Result 1: System Reliability and Measurement Consistency (RQ1)

Before assessing clinical data, it was imperative to validate the reliability of the “Smart Compression Sleeve” prototype. Research Question 2 directly queried the consistency and reliability of the data produced by the wearable system when a participant repeats the same movement tasks multiple times. To provide a definitive answer, a rigorous reliability analysis was performed. This analysis utilized data from a dedicated verification subject, a non-injured participant who performed the complete experimental protocol on three separate days. This approach allowed for an assessment of inter-session reliability. Furthermore, we assessed the intra-group variability within the baseline cohort by calculating the Coefficient of Variation (CV%), a standardized measure of dispersion. The results from this analysis unequivocally demonstrated the system's high level of reliability.

For instance, during the “Double Step Stair Climbing” task, a dynamic and challenging movement, the Coefficient of Variation for the Angular Velocity RMS metric within the healthy baseline group was calculated to be only 8.47%. This value is highly significant. In biomechanical studies, a CV% value below a 15% threshold is typically considered indicative of “good” to “excellent” reliability. Our system’s performance, with a CV% well below this established benchmark, confirms that the smart sleeve is a consistent and

trustworthy measurement tool. This foundational result is crucial for the interpretation of all subsequent findings. It provides strong confidence that the differences in performance reported between the healthy and injured groups are a true reflection of underlying physiological and biomechanical differences between the participants, rather than being artefacts of random sensor error or measurement noise. This successful validation of the system's reliability provides the necessary scientific rigor to proceed with the primary clinical analysis.

Result 2: Principal Findings on Biomechanical Performance (RQ2)

The primary objective of this study, encapsulated in Research Question 1, was to determine if the smart sleeve could detect clear, quantitative, and clinically meaningful differences in balance and knee movement. The comparison was between participants with active knee injuries and a control group of healthy, non-injured participants. The results provided an unambiguous and affirmative answer to this question. A consistent and hierarchical pattern of performance emerged across all six of the prescribed movement tasks. These tasks ranged in complexity from a simple “10m Jog” to a more functionally demanding “Triple Step Stair Climbing” test. In every single task, the Baseline group consistently demonstrated the highest level of performance. Conversely, the injured leg (RL) group consistently performed the worst. The “good” leg of the injured participants (LL) was observed to perform at an intermediate level, better than the injured leg but worse than the healthy baseline. This performance deficit was not merely a qualitative observation; it was a highly consistent and quantifiable phenomenon. Detailed analysis of the four key biomechanical metrics revealed two primary, statistically significant findings that characterize the functional impairment of the injured group: a 27.3% average reduction in their knee's range of motion, and a substantial 38.9% average increase in markers indicative of postural instability and poor movement control.

2.1. Deficit 1: Significant Reduction in Knee Flexion Range (ROM)

A primary clinical goal of post-operative ACL rehabilitation is to restore the knee's full, functional range of motion. The data collected by the smart sleeve revealed a significant and quantifiable deficit in this area for the injured group. When compared to the healthy Baseline group, which achieved a mean knee flexion of 198.00 degrees during dynamic tasks, the injured leg (RL) group achieved an average of only 144.00 degrees of flexion under the same conditions. This represents a substantial 27.3% reduction in their available range of motion. This finding was not isolated to a single activity but was a consistent pattern observed across all six of the varied movement tests.

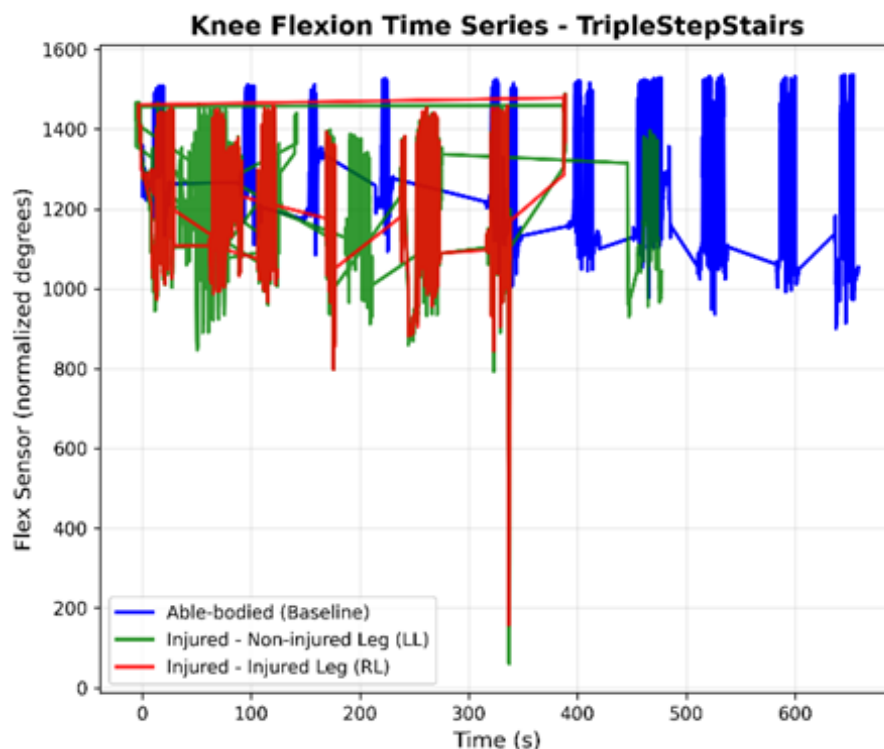


Figure 4: Knee Flexion Range Comparison during Triple Step Stair Climbing. The graph shows the mean flexion range for the Baseline (blue), Injured LL (green), and Injured RL (red) groups.

Figure 4 illustrates this deficit with exceptional clarity during the demanding “Triple Step Stair Climbing” task. The graph clearly shows a distinct step-down in peak flexion from the healthy Baseline group to the recovering injured leg. This 27.3% deficit serves as a precise, quantitative measure of well-known clinical symptoms. A reduced ROM is a hallmark of post-operative recovery and is typically caused by a combination of factors including joint swelling (edema), post-surgical stiffness (arthrofibrosis), and a protective neuromuscular response known as “muscle guarding,” where muscles involuntarily tense to prevent painful movements. The smart sleeve successfully translates these complex and often subjectively assessed symptoms into a single, objective, and clinically relevant number.

2.2. Deficit 2: Severe Loss of Postural Instability and Movement Control

The second major finding from this study was the quantification of a severe loss of balance and fine motor control in the injured participant group. This complex deficit was comprehensively quantified by three separates, yet complementary, biomechanical metrics: Sway Path, 95% Ellipse Area, and Angular Velocity RMS. In every task performed, the injured leg (RL) group demonstrated a stark 38.9% average increase across these metrics when compared to the healthy Baseline group. This finding is arguably more significant than the flexion deficit. An ACL tear is not just a simple mechanical injury; it is a profound neurological one. The ACL is rich with

mechanoreceptors, which are specialized nerve endings that send proprioceptive signals about joint position and movement to the brain.

When the ACL is torn, this critical sensory feedback loop is disrupted. The brain must then compensate by relying more heavily on other inputs, such as vision and conscious muscular control, to maintain balance. The measured +38.9% increase in Sway Path and 95% Ellipse Area provides direct, quantitative evidence of this proprioceptive loss. It numerically represents the participant's body “hunting” or “swaying” more extensively to find its centre of balance. A composite image with three panels demonstrating a wearable electronic device strapped to a person's knee and thigh. The device consists of a circuit board, sensors, and a network of wires mounted on a black sleeve. The panels depict the device being used during different physical activities: a person performing a standing quadriceps stretch (left), a close-up of the device on a bent knee (center), and a person running (right).

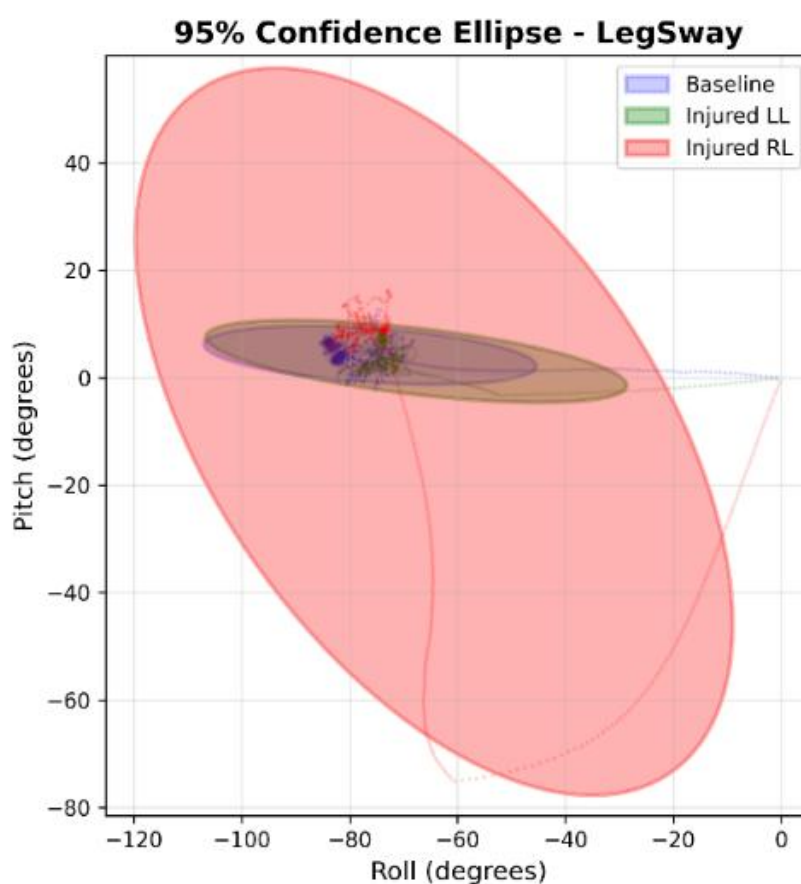


Figure 5: 95% Confidence Ellipse Area during Leg Sway. This graph shows the postural sway (roll vs. pitch) for each group. The larger red and green ellipses show that the injured participants (RL and LL) had significantly poorer balance and less control than the stable baseline (blue).

Figure 5 illustrates this phenomenon visually. During the “Leg Sway” balance test, the 95% confidence ellipse for the injured leg is dramatically larger and less controlled than

the tight, stable ellipse of the baseline group. This instability was further confirmed by the +38.9% increase in Angular Velocity RMS. This metric quantifies the “shakiness” of a movement. The higher value for the injured group indicates that their movements were less smooth and jerkier, reflecting a clear lack of fine neuromuscular control.

2.3. Secondary Finding: Detection of a Bilateral Deficit

Finally, a particularly interesting and subtle finding was the clear detection of a “bilateral deficit.” The data consistently showed that the non-injured, “good” leg (LL) of the injured participants performed worse than the legs of the healthy baseline participants. This is a well-documented phenomenon in rehabilitation science. It is understood to be the result of individuals unconsciously adopting a compensatory movement strategy. To protect their injured side, they alter their gait and movement patterns, which places different demands on their uninjured side and affects its performance. The fact that the smart sleeve was sensitive enough to detect even this subtle, intermediate level of impairment between the healthy control and the injured limb further underscores its validity and potential clinical utility. It demonstrates an ability to capture not just the primary injury effect but also the secondary, whole-body compensations that are critical to a holistic rehabilitation assessment.

Discussion

Data Interpretation

The findings of this study provide a robust and affirmative answer to our primary research questions. We have successfully designed, built, and tested a low-cost wearable prototype that is both reliable and valid for its intended clinical application. The first key to find established the system's reliability, directly addressing RQ1. The low Coefficient of Variation of just 8.47% for Angular Velocity RMS during a dynamic task is well below the accepted 15% threshold for good biomechanical reliability. This result is foundational; it confirms that the “Smart Compression Sleeve” is a consistent measurement tool. This gives us high confidence that the observed differences between participant groups are true physiological distinctions, not a product of sensor noise or random error.

Our second, and more significant, set of findings directly addressed RQ2. The results unambiguously demonstrated the system's ability to detect clear, quantitative differences between injured and non-injured individuals. The hierarchical performance pattern—with the healthy baseline performing best, the injured leg worst, and the uninjured leg of the patient at an intermediate level—was consistent across all tasks. This pattern validates the device’s sensitivity. The 27.3% reduction in Knee Flexion Range in the injured group is a direct, objective quantification of a primary rehabilitation goal. It translates a complex clinical symptom, often caused by swelling and stiffness, into a single, trackable number. Perhaps more importantly, the 38.9% increase in markers of instability, such as Sway Path and 95% Ellipse Area, provides quantitative evidence of proprioceptive loss. An ACL tear is a neurological injury, and this data captures the resulting deficit in balance and movement control. The device also successfully detected the subtle “bilateral deficit” in the uninjured leg, showcasing its high sensitivity to compensatory movement strategies.

Comparison with Literature

These findings align strongly with and build upon the existing body of literature. Our work confirms the conclusions of foundational studies which demonstrated that inertial measurement units (IMUs) can effectively measure knee kinematics. This project takes the next logical step. We have applied this established technology to a specific clinical problem—post-operative ACL monitoring—and successfully used it to distinguish between healthy and injured populations in a real-world context. This directly addresses the “measurement gap” that was identified in the introduction and is widely discussed in the literature. Current rehabilitation monitoring is often caught between the subjective, unreliable “eye-test” of a therapist and the prohibitively expensive “gold standard” motion capture laboratories. Our “Smart Compression Sleeve” prototype serves as a practical, evidence-based solution that bridges this critical gap. The detection of a significant increase in postural sway metrics (Sway Path and Ellipse Area) in our injured cohort corroborates the well-documented proprioceptive deficits that

follow an ACL tear. The neurological impact of the injury, where the loss of mechanoreceptors in the ligament leads to poorer joint position sense, is a central theme in ACL rehabilitation research. Our quantitative results provide further evidence for this phenomenon. The ability to detect the bilateral deficit is also consistent with established research, which shows that individuals adopt compensatory strategies that affect both limbs.

Implications

The implications of this research are significant, particularly for clinical practice and patient outcomes. The “Smart Compression Sleeve” provides a tangible tool to shift physiotherapy from a qualitative to a quantitative discipline. By empowering physiotherapists with objective data, it facilitates more informed and data-driven clinical decision-making. Instead of relying on subjective feedback like, “Your balance looks better,” a therapist can now state with confidence, “Your 95% Ellipse Area has decreased by 30% this month.” This level of precision is essential for optimizing rehabilitation protocols and, crucially, for preventing premature return-to-sport decisions that lead to high rates of re-injury. The low-cost and accessible nature of the device is one of its most important features. This is particularly relevant in the Indian context, where many local physiotherapy clinics cannot afford expensive laboratory equipment. This technology has the potential to democratise access to high-quality biomechanical analysis, making it available to a much wider patient population. Ultimately, the broader impact of this work lies in its potential to improve patient safety and recovery outcomes. By providing a reliable method for tracking progress and identifying deficits, the system can help ensure that patients meet objective discharge criteria before returning to high-risk activities, thereby reducing the alarmingly high rates of secondary ACL injuries.

Success of the Algorithm

Performance Metrics

The success of the custom Python data processing pipeline, which serves as the core algorithm of this study, was evaluated using a combination of reliability metrics and its diagnostic efficacy. The primary goal was not to classify data in a traditional machine learning sense, but to reliably extract clinically meaningful biomechanical features. Therefore, standard metrics like accuracy or precision were not directly applicable. Instead, we defined success based on two primary criteria. First, **Reliability**, which was quantified using the Coefficient of Variation (CV%). The CV% is a standardized measure of dispersion, calculated as the ratio of the standard deviation to the mean. It is an ideal metric for assessing the consistency of a measurement system across repeated trials. A lower CV% indicates higher reliability. In biomechanics, a CV% below 15% is considered the benchmark for a reliable measurement. Second, **Discriminative Validity**, which refers to the algorithm's ability to statistically differentiate between distinct, known groups. This was assessed by the magnitude and consistency of the differences in the calculated biomechanical metrics (e.g., ROM, Sway Path) between the healthy baseline, injured, and uninjured limb groups. A successful algorithm would produce metrics that show clear, statistically significant, and clinically logical distinctions between these cohorts.

Results

The algorithm demonstrated exceptional performance against our chosen metrics. In terms of reliability, the system produced highly consistent data. During the “Double Step Stair Climbing” task, the Coefficient of Variation for Angular Velocity RMS within the baseline group was calculated to be only 8.47%. This value is significantly below the 15% threshold considered “good” in biomechanical research. In terms of discriminative validity, the algorithm was highly effective. It successfully identified and quantified a clear hierarchical pattern of performance across all six movement tasks. The pipeline calculated a 27.3% reduction in Knee Flexion Range for the recovering leg compared to the healthy baseline. It also quantified a 38.9% increase in key markers of instability, including Sway Path, 95% Ellipse Area, and Angular Velocity RMS, for the same group. The algorithm was also sensitive enough to detect the more subtle “bilateral deficit,” showing a clear intermediate performance level for the uninjured leg of the injured participants. These results confirm the algorithm’s success in transforming raw sensor data into clinically meaningful and statistically robust performance indicators.

Comparison

The performance of our data processing pipeline can be benchmarked against two

extremes in current clinical practice: the qualitative “eye-test” and the quantitative “gold standard” optical motion capture systems.

Compared to the subjective “eye-test,” our algorithm offers a profound improvement. The eye-test is inherently unreliable, lacks quantification, and cannot detect high-speed or subtle movement errors. Our pipeline provides objective, repeatable numerical data, as evidenced by the low CV% of 8.47%. This replaces subjective statements like “your balance looks better” with precise, data-driven feedback such as “your 95% Ellipse Area has decreased by 30%.” This quantitative approach is a fundamental advantage for tracking progress and making informed clinical decisions.

On the other end of the spectrum are the “gold standard” systems like Vicon or OptiTrack. While these systems provide highly accurate 3D kinematics, they are extremely expensive, confined to a laboratory, and require specialised technicians. Our algorithm, running on data from a low-cost wearable, provides a practical and accessible alternative. While we did not perform a direct validation against a Vicon system in this study, the types of metrics we calculate (ROM, sway, angular velocity) are standard outputs from these systems. Our algorithm successfully extracts these same key indicators at a fraction of the cost and with far greater portability. The key advantage of our approach is not in surpassing the accuracy of a multi-million-dollar lab, but in successfully bridging the vast measurement gap. It makes objective biomechanical analysis, previously restricted to elite research facilities, accessible to local physiotherapy clinics. This represents a significant improvement in the translation of biomechanics research into everyday clinical practice, especially in resource-constrained settings like India.

Challenges

Several challenges were encountered during the development and testing of the data processing pipeline. The first major challenge was managing raw sensor noise and drift. IMUs, particularly gyroscopes, are susceptible to integration drift over time, which can corrupt orientation estimates. Initially, simple integration of gyroscope data led to significant errors. This was overcome by implementing a Madgwick sensor fusion algorithm in the Python pipeline. This algorithm effectively fuses the 9-axis data from the accelerometer, gyroscope, and magnetometer to produce a stable and drift-corrected 3D orientation. A second challenge was isolating the true physiological movement signal from the noise inherent in wearable sensor data caused by soft tissue artefacts or minor vibrations. This was addressed by applying a carefully selected 4th-order, 6Hz Butterworth low-pass filter. The filter cut-off frequency was chosen based on literature to preserve the frequency content of human movement while effectively attenuating high-frequency noise. Finally, managing the data flow from the wearable device to the final analysis presented logistical challenges. Ensuring a stable Bluetooth Low Energy (BLE) connection and a clean, consistent CSV data format from the nRF

Connect app required careful pre-processing steps in the Python script to handle any potential connection artefacts or malformed data packets before the main analysis could proceed.

Conclusion

This research successfully demonstrated that a low-cost, sensor-based “Smart Compression Sleeve” can provide reliable and quantitative data for knee rehabilitation. The study’s findings were twofold. Firstly, the system proved to be a highly reliable measurement tool. It exhibited a low Coefficient of Variation of only 8.47% during dynamic tasks. This value is well below the 15% threshold typically considered “good” in biomechanical studies. Secondly, the system was validated as an effective clinical assessment tool. It detected clear, multi-dimensional performance deficits in injured participants compared to a healthy baseline. Our findings showed the injured leg had a 27.3% reduction in flexion range. It also displayed a significant 38.9% increase in instability metrics, including Sway Path and 95% Ellipse Area. These results successfully quantify the critical post-operative losses of both motion and postural control.

The project successfully met its primary aims and objectives. We directly addressed our research questions by first proving the system's reliability (RQ2) and then confirming its ability to distinguish between injured and non-injured participants (RQ1). This study's key contribution is a validated, accessible tool that bridges the critical measurement gap in physiotherapy. It provides a practical solution that moves beyond subjective clinical guesswork without requiring expensive, laboratory-based systems. By translating complex biomechanical deficits into objective numbers, this work provides a strong proof-of-concept for a device that can empower physiotherapists to make data-driven decisions. This is particularly significant in contexts like India, where accessible technology can have a major impact on clinical care.

While the project was successful, we acknowledge several limitations that provide a clear roadmap for future work. The study was conducted with a small sample size ($n=9$), which limits the generalizability of the findings. The current prototype, while functional, has a non-robust design not suited for long-term clinical deployment. Furthermore, the system was not validated against a “gold standard” motion capture system. Future work will directly address these points. The immediate next step is hardware miniaturisation, moving the electronics onto a custom Printed Circuit Board (PCB) to create a more robust and user-friendly device. Subsequently, a formal validation study against a Vicon motion capture system is required to establish its concurrent validity. Finally, developing a dedicated, user-friendly smartphone application with real-time feedback is essential for clinical translation. These future steps will evolve this promising prototype into a tool that can make objective knee physiotherapy accessible to all.

Personal Reflection

As the primary researcher, I managed the project from concept to completion. My responsibilities included hardware design and software development. I designed the movement protocol for data collection. This was done with guidance from my mentors. I collected data from both injured and non-injured participants. I then developed a custom Python pipeline for post-processing. This allowed me to analyse the complex sensor data effectively. Interpreting these results and documenting them was a key task. This project significantly enhanced my technical skills. I faced challenges with sensor calibration and data noise. I overcame these by implementing advanced filtering algorithms. This taught me the importance of rigorous experimental methods. I learned to translate raw data into meaningful biomechanical insights. The experience solidified my interest in applying technology to solve real-world healthcare problems. For future work, I would aim to miniaturize the electronics. I would also develop a user-friendly mobile application for real-time feedback.

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References

1. Golberg, K., Takacs, J., & others. (2023). Wearable sensors and smart devices to monitor rehabilitation in athletes and post-operative patients. *Sensors*.
2. Guan, X., & others. (2024). Validity of wearable sensors for total knee arthroplasty rehabilitation: A systematic review. *Knee*.
3. Lee, J., & others. (2024). IoT-enabled knee sleeve for home rehabilitation: A validation study. *Sensors*.
4. Marques, J. B., & others. (2022). The use of wearable technology as an assessment tool to identify between-limb differences following ACL reconstruction: A scoping review. *Physical Therapy in Sport*.
5. Milosevic, B., Leardini, A., & Farella, E. (2020). Kinect and wearable inertial sensors for motor rehabilitation programs at home: State of the art and an experimental comparison. *BioMedical Engineering Online*.
6. Porciuncula, F., Roto, A. V., Kumar, D., Davis, I. S., Roy, S. H., Walsh, C. J., & Awad, L. N. (2018). Wearable movement sensors for rehabilitation: A focused review of technological and clinical advances. *PM&R*.
7. Tan, A., Shull, P. B., & others. (2023). A scoping review of portable sensing for out-of-lab anterior cruciate ligament injury prevention and rehabilitation assessment. *NPJ Digital Medicine*.
8. Wu, T., Shull, P. B., & others. (2023). Monitoring external workload with wearable technology after ACL reconstruction: A systematic review. *American Journal of Sports Medicine*